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Identification of Dynamic Discrete Choice Models

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discrete decision process, optimal stopping, hazard, hitting time, heterogeneity

Abstract

Econometric models of dynamic discrete choice processes are applied to a wide variety of economic problems. Recent research on their empirical content has brought important new insights. It has clarified the conditions for their identification from choice and covariate panel data in the absence of dynamic selection on unobservables. It has provided important new identification results for discrete-time models with unobserved heterogeneity and unobserved states. Finally, it has enhanced the attractiveness of continuous-time models, by developing new insights on the identification of continuous-time optimal stopping models. Current developments in the literature promise to shed further light on the specification and identification of models with unobserved state variables, theory-based non-proportional hazard models, continuous-time optimal stopping models with time-varying covariates, and dynamic games in discrete and continuous time.

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1. INTRODUCTION

Empirical economists now commonly use econometric models that explicitly specify agents' dynamic discrete choices as the solution to dynamic programming problems. Broadly, three classes of models can be distinguished: (*a*) discrete-time models, (*b*) continuous-time models in which agents take discrete decisions at random and discrete (Poisson) times, and (*c*) continuous-time models driven by Brownian motion or more general persistent processes.

Discrete-time dynamic programming models have been applied to a wide range of economic problems. Keane & Wolpin (2009) review examples of recent empirical studies of schooling, employment, fertility, and health insurance choices. Advances in the econometric analysis of dynamic discrete games, such as firm entry and exit games in industrial organization, have further amplified their relevance because, in these games' Markov perfect equilibria, each agent solves a dynamic program given the other players' strategies (see Ackerberg et al. 2007 for a review).

Heckman & Singer (1985, 1986) present examples of continuous-time models based on Poisson processes. Among these, sequential job search models are now a key tool in the empirical analysis of labor market dynamics (Eckstein & Van den Berg 2007). In them, job offers arrive at Poisson times, and agents decide whether to accept job offers when they have arrived. Typically, they use threshold rules: Agents accept jobs that offer a wage above a certain reservation wage. Similar models have appeared in insurance economics in which agents decide on claiming losses that are incurred at Poisson times (Abbring et al. 2008). The key distinguishing feature of this class of models is that transitions only take place at the Poisson times of the shocks. This naturally leads to a hazard specification in terms of the intensity of those shocks (job-offer arrival rate) and the conditional probabilities that these trigger transitions (job acceptance probabilities). Consequently, these models can be analyzed with standard hazard methods.

Continuous-time models driven by Brownian motion, and more general persistent processes, are central to the options literature in finance, and its applications to real options problems (Dixit & Pindyck 1994, Stokey 2009). In these models, agents typically make optimal transitions when the latent process first hits a threshold. Shimer's (2008) analysis of unemployment durations is one recent example. They share this feature with many of the models discussed in this review. The models that we classify as hitting-time models stand out because their latent processes creep across (attain), not jump across, the threshold at the first hitting time. This ensures that standard methods for the analysis of hitting times of Brownian motion can be applied or extended to analyze these models. Hazard specifications only follow indirectly.

All three classes of models are applied with the ultimate ambition to learn about the primitives of dynamic decision processes. If these primitives can be measured from data, the effects of counterfactual policy experiments can be computed, state dependence and dynamic selection can be separated, and risk and heterogeneity can be disentangled. This review addresses the fundamental question of to what extent it is possible to determine model primitives from choice, outcome, and covariate panel data.

The discrete- and continuous-time literatures have much in common. Both usually employ Markov decision processes. Some of the early applications of discrete-time models, such as Pakes's (1986) study of patent renewal and values and Rust's (1987) analysis of bus-engine renewal, involved optimal stopping problems similar to those in the

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continuous-time real options literature. Empirical job search models have been implemented in both discrete and continuous time, and Flinn & Heckman (1982) explicitly note that their identification results for continuous-time search models can be applied in discrete time. However, many fields of economics have settled on either discrete- or continuous-time models, and many of the techniques for their analysis differ substantially. Consequently, both literatures have mostly developed separately. In this review, we bring them together, in the hope that this inspires progress both ways.

This review updates and complements earlier reviews. Rust's (1994) key review of discrete-time Markov discrete choice models contains his well-known nonidentification result but necessarily misses more recent developments. Aguirregabiria & Mira's (2010) extensive update focuses on specification and estimation of discrete-time models and ignores identification. Van den Berg (2001) discusses the specification and identification of continuous-time mixed proportional hazards (MPH) models, and Eckstein & Van den Berg (2007) review empirical search models. We complement these papers with a brief, unifying update of the literature on the MPH model and a review of recent progress in applying this literature to hitting-time models. Finally, Abbring & Heckman (2007, 2008) review the literature on dynamic treatment effects and dynamic policy evaluation. We focus more narrowly on dynamic discrete choice models and discuss some of the most recent developments.

The article is organized as follows. Section 2 reviews identification results for discrete-time models. Sections 3 and 4 discuss continuous-time hazard and hitting-time models. Section 5 concludes with some reflections on the state of the literature.

2. DISCRETE-TIME MODELS

This section reviews the identifiability of discrete decision processes in discrete time from panel data on choices, state variables, outcomes, and auxiliary measurements. In Section 2.1, we first present a basic dynamic discrete decision model with properties shared by much, but not all, of the analysis reviewed in this section: Markov state dynamics, rational beliefs, and time-separable preferences.

Section 2.2 reviews identification of the simplest econometric implementation of this framework, based on Rust's (1987) conditional independence assumption. This assumption, with the assumption that utility is additively separable in observables and unobservables, facilitates a discussion of the identification of primitives, without needing to worry about dynamic selection on unobservables. Much of the recent research on dynamic discrete choice analysis and dynamic discrete games is firmly rooted in Rust's framework: Either it relies directly on his conditional independence assumption or it uses results for his framework in a second stage, after dealing with selection on observables in a first stage.

Section 2.3 reviews an example of the latter approach, by discussing the first-stage identification of models with discrete unobserved heterogeneity. Such models have been popularized by the work of Eckstein & Wolpin (1990, 1999), Keane & Wolpin (1997), and others and are arguably the most common extension of Rust's framework with nontrivial dynamic selection.

Sections 2.4 and 2.5 discuss recent developments in the identification of models with more general serially correlated unobservables. In both sections, identification relies crucially on observations of both choices and choice-specific outcomes, and on cross-equation restrictions dictated by economic theory.

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2.1. Markov Discrete Decision Processes

We first introduce the basic Markov dynamic discrete decision problem that is central to the econometric dynamic discrete choice literature reviewed in this section.

2.1.1. Setup. Consider an agent living in discrete time, indexed by $t \in \mathcal{T}^*$. The agent's horizon may be finite, in which case $\mathcal{T}^* \equiv \{1, \ldots, \bar{T}^*\}$ for finite $\bar{T}^*$. It may also be infinite; then, $\mathcal{T}^* \equiv \{1, 2, \ldots\}$.

At each time t, a state vector S_t, with support $\mathcal{S}$, is revealed to the agent. Subsequently, the agent chooses an action from a set $\mathcal{A}_t^*(S_t) \subseteq \mathcal{A} = \{1, 2, \ldots, J\}$, with $1 \leq J < \infty$. The agent's choice of action affects both current payoffs directly, and future payoffs indirectly, through its controlling effect on future states. Specifically, if the agent selects action $a \in \mathcal{A}_t^*(S_t)$, he collects utility $u_t^*(a, S_t)$ and, if $t < \bar{T}^*$, moves to period $t+1$ with a new state vector S_{t+1} drawn, independently from states and actions before period t, from the (controlled) transition distribution $Q_{t+1}^*(\cdot | a, S_t)$.

The agent chooses actions that maximize his expected utility, discounted with a factor $0 \leq \rho < 1$. He has rational beliefs and uses the objective Markov transition distributions Q_{t+1}^* to form his expectations over future states and decisions.

2.1.2. Solution. The solution to this agent's Markov discrete decision process can be analyzed recursively. Denote the agent's optimal action at time t with A_t. Under standard regularity conditions, A_t satisfies

$$A_t = \arg \max_{a \in \mathcal{A}_t^*(S_t)} U_t^*(a, S_t), \tag{1}$$

where U_t^*, $t \in \mathcal{T}^*$, solves the Bellman equation

$$U_t^*(a, S_t) = u_t^*(a, S_t) + \rho \int \max_{a' \in \mathcal{A}_{t+1}^*(s')} U_{t+1}^*(a', s') dQ_{t+1}^*(s' | a, S_t), \; t < \bar{T}^*. \tag{2}$$

If the horizon is finite, under regularity conditions, Equation 2 can be solved backward, from a terminal condition $U_{\bar{T}^*}^* = u_{\bar{T}^*}^*$. Then, there exists a Markov decision rule $\alpha_t : \mathcal{S} \to \mathcal{A}$ such that $A_t = \alpha_t(S_t)$ satisfies Equation 1, $t \in \mathcal{T}^*$.

With an infinite horizon, the agent's problem is particularly easy to solve if it is stationary, that is, if none of its primitives depends on time. Then, under standard regularity conditions, contraction arguments imply that the Bellman operator implicitly defined by the right-hand side of Equation 2 has a unique fixed point U^*, so that Equation 1 defines a stationary Markov decision rule α.

2.2. Conditional Independence

Rust (1987, 1994) empirically implements the previous section's Markov decision framework by distinguishing between state variables X_t that are observed by the econometrician and state variables W_t that are not and by limiting the role of the unobserved state variables, so that they do not help the agent in predicting future payoffs and do not lead to dynamic econometric selection problems.

2.2.1. Setup. Let $S_t = (X_t, W_t)$, where X_t takes values in $\mathcal{X}$, and $W_t \equiv [W_t(1), \ldots, W_t(J)]$ is continuously distributed on $\mathbb{R}^J$. Rust differentiates their roles in the Markov discrete

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decision model in three ways. (*a*) Choice sets only depend on the observed state variables, so that $\mathcal{A}_t^*(X_t, W_t) = \mathcal{A}_t(X_t)$. (*b*) Utility is additively separable in a general component depending on observed state variables and actions and a choice-specific unobserved state shock: $u_t^*(a, X_t, W_t) = u_t(a, X_t) + W_t(a)$. (*c*) The controlled Markov transition distribution Q_{t+1}^* satisfies a conditional independence condition: Its density (relative to an appropriate dominating measure) q_{t+1}^* can be decomposed as

$$q_{t+1}^*(X_{t+1}, W_{t+1} \mid a_t, X_t, W_t) = q_{t+1}(X_{t+1} \mid a_t, X_t) g_{t+1}(W_{t+1} \mid X_{t+1}), \tag{3}$$

for some density q_{t+1} and a Lebesgue density g_{t+1}.

The conditional independence assumption in Equation 3 embodies two restrictions: (*a*) The observed state variables evolve, independently from the unobserved state variables, as a controlled first-order Markov process, and (*b*) the unobserved state variables evolve independently from the agent's actions and only depend on the state's history through the concurrent observed state variable. This excludes both permanent unobserved heterogeneity and autocorrelated unobserved state variables. Restriction (*a*) implies that the evolution of the observed state variables is not informative on the unobservables; this distinguishes these state variables from the choice-specific outcomes considered in Sections 2.4 and 2.5.

For completeness, we also decompose the density q_1^* of the initial state as $q_1^*(X_1, W_1) = q_1(X_1) g_1(W_1 | X_1)$, for some density q_1 and Lebesgue density g_1. We furthermore set $\mathbb{E}[W_t \mid X_t] = \int w g_t(w \mid X_t) dw = 0$, so that $\mathbb{E}\left[u_t^*(a, X_t, W_t) \mid X_t\right] = u_t(a, X_t)$. In fact, in applied work, it is often assumed that the unobserved state variables W_t are simply independent shocks.

These separability and conditional independence assumptions facilitate the analysis of Equations 1 and 2 on a smaller state space. In particular, they imply that

$$A_t = \alpha_t(X_t, W_t) = \arg \max_{a \in \mathcal{A}_t(X_t)} \{U_t(a, X_t) + W_t(a)\}, \tag{4}$$

where the choice-specific (expected) value function $U_t(a, X_t) \equiv U_t^*(a, X_t, W_t) - W_t(a)$ solves the Bellman equation

$$U_t(a, X_t) = u_t(a, X_t) + \rho \int R_{t+1}(x') dQ_{t+1}(x' \mid a, X_t). \tag{5}$$

Here,

$$R_{t+1}(x') \equiv \int \max_{a' \in \mathcal{A}_{t+1}(x')} \left\{U_{t+1}(a', x') + w'(a')\right\} g_{t+1}(w' \mid x') dw'$$

is McFadden's (1981) social surplus function.

2.2.2. Identification. Now consider identification. Suppose that we observe the discrete choices A_t and state variables X_t for $t \in \mathcal{T} \equiv \{1, \ldots, \bar{T}\}$, with $2 \leq \bar{T} \leq \bar{T}^*$ finite. We abstract from sampling error and assume that the data provide us with the joint distribution of $[(A_1, X_1), \ldots, (A_{\bar{T}}, X_{\bar{T}})]$. Note that this distribution is fully determined by the terminal choice-specific values $U_{\bar{T}}$ and the following primitives: (*a*) the initial distribution Q_1 of X_1; (*b*) the Markov transition distributions Q_{t+1}, $t \in \{1, \ldots, \bar{T} - 1\}$; (*c*) the distributions G_t of the choice-specific shocks, $t \in \mathcal{T}$; (*d*) the utility functions u_t, $t \in \{1, \ldots, \bar{T} - 1\}$; and (*e*) the discount factor ρ. Consequently, at best we can hope to identify this subset of primitives, and $U_{\bar{T}}$, from the data. In the special case that $\bar{T} = \bar{T}^*$, we observe the full choice history, and this

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includes all primitives, in particular, $u_{\bar{T}} = U_{\bar{T}}$. In the stationary infinite-horizon case, the subset of primitives itself includes all primitives and uniquely determines $U_{\bar{T}}$. In general, we need to treat $U_{\bar{T}}$ as a primitive to be identified (Magnac & Thesmar 2002).

The initial state distribution Q_1 is formally a primitive of the decision problem because we have implicitly assumed that the data are collected from this problem's initial period onward. If we relax this assumption, Q_1 will be the distribution of the observed state some time after its initialization. In the stationary case, we could take this to the limit and interpret Q_1 in terms of the other primitives, as the long-run (ergodic) distribution of X_t.

We would like to know whether the data on $[(A_1, X_1), \ldots, (A_{\bar{T}}, X_{\bar{T}})]$ uniquely determine the relevant primitives and $U_{\bar{T}}$. Throughout, it should be understood that identification can only be obtained on the relevant supports. Moreover, for expositional convenience only, suppose that $\mathcal{X}$ is discrete, with the densities q_t giving probabilities (that is, they are densities relative to the counting measure).

First note that, by virtue of Rust's (1987) assumptions, the density $q_1(x) = \Pr(X_1 = x)$ and the transition density $q_{t+1}(x|a, X_t) = \Pr(X_{t+1} = x | A_t = a, X_t)$ (for $t \in \{1, \ldots, \bar{T} - 1\}$) are directly identified from the data. Moreover, for $t \in \mathcal{T}$, the conditional choice probability $p_t(a|X_t) \equiv \Pr(A_t = a|X_t)$ identifies an average over $\alpha_t(X_t, W_t)$ that is free from dynamic selection bias and that coincides with the probability that an individual agent would attach to choosing a in period t if she would know X_t but not yet W_t. Finally, the density f of the data conveniently factorizes as

$$f[(A_1, X_1), \ldots, (A_{\bar{T}}, X_{\bar{T}})] = q_1(X_1)p_1(A_1 \mid X_1) \prod_{t=2}^{\bar{T}} q_t(X_t \mid A_{t-1}, X_{t-1})p_t(A_t \mid X_t). \quad (6)$$

This suggests that the conditional choice probabilities can be used to identify U_t and G_t using results on the identification of static discrete choice models. A simple example clarifies this point. Suppose that the components of W_t are independently distributed with type I extreme value distributions. This fixes G_t and implies that the conditional choice probabilities have the well-known multinomial logit form:

$$p_t(a \mid X_t) = \frac{\exp[U_t(a, X_t)]}{\sum_{b \in \mathcal{A}_t(X_t)} \exp[U_t(b, X_t)]}, \quad a \in \mathcal{A}_t(X_t).$$

Thus, for any two actions $a, a^* \in \mathcal{A}_t(X_t)$, the log odds ratio $\ln[p_t(a|X_t)/p_t(a^*|X_t)]$ identifies the choice-specific value contrast $U_t(a, X_t) - U_t(a^*, X_t)$, and this is all the information that can possibly be pulled from the conditional choice data. Hotz & Miller (1993) show that this inversion of the conditional choice probabilities for the logit case generalizes: For any given G_t, type I extreme value or other distribution, choice-specific value contrasts are uniquely determined by the choice probabilities.

Now, as pointed out by Rust (1994), this is not sufficient for identification: If $U_t(a, X_t)$ produces the choice probability data, then so does $\tilde{U}_t(a, X_t) \equiv U_t(a, X_t) + h_t(X_t)$. In Rust's stationary, infinite-horizon case, this corresponds to a particular transformation of the utility function,

$$\tilde{u}(a, X_t) = u(a, X_t) + h(X_t) - \rho \int h(x')q(x' \mid a, X_t)dx'.$$

This is a manifestation of the standard problem in static discrete choice analysis that only utility contrasts are identified.

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Even if we settle this problem by fixing the choice-specific value in one of the alternative choices, for each possible state in $\mathcal{X}$, the model is not identified. For expositional convenience, suppose that $\mathcal{A}_t(X_t) = \mathcal{A}$, and set $U_t(J, \cdot) = 0$. Then $U_t(a, X_t)$ is identified, but $u_t(a, X_t)$ is not. In particular, whatever the underlying primitives really are, a static model, with discount factor 0 and utility function $\tilde{u}_t = U_t$, rationalizes the choice data (Manski 1993, Rust 1994).

Magnac & Thesmar (2002) make the degree of underidentification of the model under conditional independence precise by showing that, in general, fixing ρ, G_t, and the utilities u_t and terminal value $U_{\bar{T}}$ in one (feasible) reference alternative is just enough for identification.

From Equation 5, it is clear that exclusion restrictions on the utilities u_t facilitate the identification of ρ. Fix G_t and assume that U_t is identified. It is shown above that this is not sufficient to identify u_t and ρ. Now suppose that a state variable is excluded from u_t but affects the choice-specific values through its effect on the transitions of the other state variables and forward-looking preferences ($\rho > 0$). Then, the corresponding choice data cannot be rationalized with a static model ($\rho = 0$). Stationarity of utilities can be used as an exclusion restriction: If u_t does not depend on t, but Q_{t+1} does, then time variation in choice-specific values can be used to pin down ρ.

By itself, without time variation in Q_{t+1}, a stationarity restriction does not break the nonidentification of ρ. It does, however, avoid the need to specify the terminal value: Bajari et al. (2009) prove identification for the stationary, infinite-horizon case given that ρ, G, and the utility u in one reference alternative are fixed.

In empirical research, functional form restrictions are common sources of identification. A specific example is the extreme value specification of G_t. More generally, identification results like Matzkin's (1992) for static discrete choice models suggest that G_t and choice-specific value contrasts can be identified nonparametrically provided these functions satisfy certain shape restrictions.

Finally, data on choices and state variables are sometimes complemented with information on choice-specific outcomes and other measurements that are informative on the unobservables. Restrictions imposed by economic theory across choices and outcomes can be powerful sources of identification. Sections 2.4 and 2.5 provide further discussion and examples.

2.2.3. Relaxing conditional independence. In many economic applications, agents are likely to have better information about their futures than the econometrician. If they act on this superior information, then Rust's (1987) conditional independence assumption typically breaks down. This motivates augmenting this section's framework with unobserved persistent state variables. In such an augmented framework, the observed history of the state variables and actions not only causally determines the agent's current state, but is also statistically informative on the agent's unobserved persistent state. Consequently, the observed state and choice dynamics confound agent-level state dependence with dynamic selection effects (Heckman 1981).

Before we review recent attempts to address this problem in the next few sections, it is useful to recall why this is difficult. To this end, consider the observational implications of the conditional independence assumption. It is immediately clear from the factorization of the data density in Equation 6 that, together with the Markov assumption, conditional independence substantially restricts time-series variation in the data. However, in isolation from the Markov restriction, conditional independence has little bite: If we allow for full (non-Markov) dependence of observed states and choices on their histories, we can match

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any given data distribution without dynamic selection on unobservables (see also Abbring & Heckman 2007, sections 3.2.1.2 and 3.2.2.2). This underscores the key role of the Markov restriction, combined with stationarity and other restrictions, in the identification of models with unobserved persistent states.

2.3. Discrete Unobserved Heterogeneity

The simplest way to endow agents with better information than the econometrician is to allow for permanent differences between agents of which the agents are aware, but that the econometrician does not observe. To this end, augment the state variables of the previous section's model with a permanent unobserved heterogeneity component V: $S_t = (X_t, W_t, V)$, where, as before, X_t and W_t are time-varying observed and unobserved state variables.

We assume that V has a discrete distribution, with finite support $\{v_1, v_2, \ldots, v_K\}$. This specification has been popular since it appeared in Heckman & Singer's (1984b) nonparametric estimator of the MPH model. It has been promoted and used in the context of dynamic discrete decision processes by Eckstein & Wolpin (1990, 1999) and Keane & Wolpin (1997), among others.

2.3.1. Mixture setup and identification. All primitives, including Q_1 and $U_{\bar{T}}$, may generally depend on V, with $u_t^*(a, X_t, W_t, V) = u_t(a, X_t, V) + W_t(a)$. For expositional convenience, we focus on binary choice and set $\mathcal{A}_t^*(S_t) = \{0,1\}$. We now assume that conditional independence of $\{X_t, W_t\}$ holds conditional on V. This ensures that a mixed version of the factorization in Equation 6 holds:

$$\begin{aligned} &f[(A_1, X_1), \ldots, (A_{\bar{T}}, X_{\bar{T}})] \\ &\quad = \sum_{k=1}^{K} \pi_k q_1(X_1 \mid v_k) p_1(A_1 \mid X_1, v_k) \prod_{t=2}^{\bar{T}} q_t(X_t \mid A_{t-1}, X_{t-1}, v_k) p_t(A_t \mid X_t, v_k), \end{aligned} \tag{7}$$

where $\pi_k \equiv \Pr(V = v_k) > 0$. Because of conditional independence, V is the only source of dynamic selection. Consequently, the components in the right-hand side of Equation 7 have structural interpretations. In particular, $q_t(X_t | A_{t-1}, X_{t-1}, v_k)$ is the true transition density of X_t for an agent with $V = v_k$, and the conditional choice probabilities $p_t(A_t | X_t, v_k)$ again identify an average over the decision rules that is free from dynamic selection bias. However, in contrast to Section 2.2, neither these components nor the initial state density q_1 are directly identified.

Kasahara & Shimotsu (2009) recently adapted and extended results on the identification of finite mixtures to variants of this framework. Their approach takes mixtures such as Equation 7 as given and explores identification of the components of its right-hand side from the data in its left-hand side. Their analysis neither requires nor exploits these components' structural interpretation; in particular, they do not use the structure of the Markov discrete decision framework. Thus, an application to this structural framework is an example of a two-stage approach to identification: First, heterogeneity is separated from structural state transition and choice components; second, the decision problem's primitives are determined from the latter, using results similar to those in Section 2.2.

2.3.2. Baseline case. We first review Kasahara & Shimotsu's baseline analysis. This analysis studies a special case of Equation 7, with the transition densities q_t, $t \in \{2, \ldots, \bar{T}\}$,

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independent of V and the conditional choice probabilities p_t independent of t. This baseline case is of particular interest to this review because it can be structurally applied. It corresponds to a stationary infinite-horizon Markov decision model with heterogeneity in static and dynamic preferences and heterogeneous initial conditions, but with homogeneous state transitions (see also Kasahara & Shimotsu 2009, example 2).

Note that Kasahara & Shimotsu in fact allow for general (non-Markov) dependence of X_t on the observed state and choice history. They also allow the initial conditional choice probability p_1 to differ from the stationary conditional choice probability p. However, neither these extensions nor, in fact, the time dependence of the state transition probabilities are consistent with a structural interpretation of the stationary conditional choice probabilities p and the initial conditions. Therefore, we ignore these extensions here.

For expositional convenience, as in the previous section and following Kasahara & Shimotsu, suppose that $\mathcal{X}$ is discrete and that the densities q_t are probabilities. In the baseline case, as in Section 2.2, the structural transition densities are identified directly by $q_{t+1}(x|a, X_t) = \Pr(X_{t+1} = x|A_t = a, X_t)$. They can also be taken outside the mixture in the right-hand side of Equation 7, so that we can focus on identification of the components of the right-hand side of

$$\begin{aligned} f^*[(A_1, X_1), \ldots, (A_{\bar{T}}, X_{\bar{T}})] &\equiv \frac{f[(A_1, X_1), \ldots, (A_{\bar{T}}, X_{\bar{T}})]}{\prod_{t=1}^{\bar{T}-1} q_{t+1}(X_{t+1} \mid A_t, X_t)} \\ &= \sum_{k=1}^{K} \pi_k q_1(X_1 \mid v_k) \prod_{t=1}^{\bar{T}} p(A_t \mid X_t, v_k) \end{aligned} \tag{8}$$

from f^*. For a given evaluation point of $(X_1, \ldots, X_{\bar{T}})$, the right-hand side is a (defective) finite discrete mixture over the product of $\bar{T}$ binary choice probabilities. The identification of such mixtures has been studied in psychometrics, biometrics, and econometrics in the 1950s (e.g., Anderson 1954). Kasahara & Shimotsu substantially improve on these results by exploiting variation with the observed state paths $(X_1, \ldots, X_{\bar{T}})$ and not only marginal sequences of binary outcomes.

The key idea in this approach is that a rich set of restrictions on the component densities and the mixing probabilities can be constructed by evaluating a range of marginal versions of Equation 8. Assume that Q_1, $Q_{t+1}(\cdot|0, X_t)$, and $Q_{t+1}(\cdot|1, X_t)$ have full support $\mathcal{X}$, almost surely. Let $\bar{T} = 3$ and $(x_1, x_2, x_3) \in \mathcal{X}$. Then, directly evaluating Equation 8 at $[(1, x_1), (1, x_2), (1, x_3)]$ gives one restriction,

$$f^{x_1}_{x_2,x_3} \equiv f^*[(1, x_1), (1, x_2), (1, x_3)] = \sum_{k=1}^{K} \pi_k q_1(x_1 \mid v_k) \prod_{t=1}^{3} p(1 \mid x_t, v_k). \tag{9}$$

First integrating out (A_1, X_1) from Equation 8 and then evaluating at $[(1, x_2), (1, x_3)]$ give another,

$$f_{x_2,x_3} \equiv \sum_{a_1=0}^{1} \sum_{x_1 \in \mathcal{X}} f^*[(a_1, x_1), (1, x_2), (1, x_3)] = \sum_{k=1}^{K} \pi_k \prod_{t=2}^{3} p(1 \mid x_t, v_k). \tag{10}$$

Similarly, we can derive a range of additional restrictions by integrating over one or more of (A_1, X_1), A_2, and A_3 and evaluating. Denote the result of integrating the left-hand side of Equation 8 over A_3 with $f^{x_1}_{x_2}$, the result of integrating over A_2 and A_3 with f^{x_1}, and so on. Then, we have

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$$
\begin{aligned}
f_{x_2}^{x_1} &= \sum_{k=1}^{K} \pi_k q_1(x_1 \mid \nu_k) p(1 \mid x_1, \nu_k) p(1 \mid x_2, \nu_k), \\
f_{x_3}^{x_1} &= \sum_{k=1}^{K} \pi_k q_1(x_1 \mid \nu_k) p(1 \mid x_1, \nu_k) p(1 \mid x_3, \nu_k), \\
f^{x_1} &= \sum_{k=1}^{K} \pi_k q_1(x_1 \mid \nu_k) p(1 \mid x_1, \nu_k), \qquad (11) \\
f_{x_2} &= \sum_{k=1}^{K} \pi_k p(1 \mid x_2, \nu_k), \text{ and} \\
f_{x_3} &= \sum_{k=1}^{K} \pi_k p(1 \mid x_3, \nu_k).
\end{aligned}
$$

Without covariates, that is, if $\mathcal{X}$ contains only one element (x_1, x_2, x_3), this gives seven restrictions, corresponding to the seven probabilities that characterize the distribution of the eight possible binary choice paths $(a_1, a_2, a_3) \in \{0,1\}^3$. In this case, because $q_1(x_1|\nu_k) = 1$ and $\sum_{k=1}^{K} \pi_k = 1$, we need to determine $3K$ conditional choice probabilities and $K - 1$ mixing probabilities. Therefore, we can identify at most the case with $K = 2$ types, which would have exactly seven unknown parameters. Indeed, Anderson (1954) gives sufficient conditions for identification if $\bar{T} \geq 2K - 1$, and others have improved on his results.

Kasahara & Shimotsu's contribution to the baseline case is to note that Equations 9–11 may yield different restrictions for different covariate paths $(x_1, x_2, x_3) \in \mathcal{X}^3$ and to develop sufficient conditions for identification with $\bar{T} = 3$ when covariate variation is available. Their analysis is intuitive and close in spirit to classical contributions such as Anderson's (1954). It is instructive to review it here.

For $x \in \mathcal{X}$ and $\xi \equiv (\xi_1, \ldots, \xi_{K-1}) \in \mathcal{X}^{K-1}$, let

$$
L_\xi \equiv \begin{bmatrix} 1 & p(1 \mid \xi_1, \nu_1) & \cdots & p(1 \mid \xi_{K-1}, \nu_1) \\ \vdots & \vdots & \ddots & \vdots \\ 1 & p(1 \mid \xi_1, \nu_K) & \cdots & p(1 \mid \xi_{K-1}, \nu_K) \end{bmatrix}
$$

be a $K \times K$ matrix with conditional choice probabilities,

$$
D_x \equiv \operatorname{diag}[q_1(x \mid \nu_1) p(1 \mid x, \nu_1), \ldots, q_1(x \mid \nu_K) p(1 \mid x, \nu_K)]
$$

a $K \times K$ diagonal matrix with initial state and choice probabilities on its diagonal, and $\Pi \equiv \operatorname{diag}(\pi_1, \ldots, \pi_K)$ a $K \times K$ diagonal matrix with mixing probabilities. These are parameters to be identified. Collect corresponding probabilities known from the data in two $K \times K$ matrices,

$$
F_\xi \equiv \begin{bmatrix} 1 & f_{\xi_1} & \cdots & f_{\xi_{K-1}} \\ f_{\xi_1} & f_{\xi_1,\xi_1} & \cdots & f_{\xi_1,\xi_{K-1}} \\ \vdots & \vdots & \ddots & \vdots \\ f_{\xi_{K-1}} & f_{\xi_{K-1},\xi_1} & \cdots & f_{\xi_{K-1},\xi_{K-1}} \end{bmatrix}
$$

and

$$
F_\xi^x \equiv \begin{bmatrix} f^x & f_{\xi_1}^x & \cdots & f_{\xi_{K-1}}^x \\ f_{\xi_1}^x & f_{\xi_1,\xi_1}^x & \cdots & f_{\xi_1,\xi_{K-1}}^x \\ \vdots & \vdots & \ddots & \vdots \\ f_{\xi_{K-1}}^x & f_{\xi_{K-1},\xi_1}^x & \cdots & f_{\xi_{K-1},\xi_{K-1}}^x \end{bmatrix}
$$

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Then,

$$F_\xi = L'_\xi \Pi L_\xi \text{ and } F^x_\xi = L'_\xi D_x \Pi L_\xi \tag{12}$$

express data in terms of the mixing components and probabilities.

Kasahara & Shimotsu note that the model can be identified if the equations in Equation 12 can be inverted for some $x \in \mathcal{X}$ and $\xi \in \mathcal{X}^{K-1}$. They give the following sufficient conditions for this: There exist (*a*) a $\xi \in \mathcal{X}^{K-1}$ such that L_ξ is full rank and (*b*) an $x \in \mathcal{X}$ such that D_k is full rank, with none of its K diagonal elements identical (Kasahara & Shimotsu 2009, proposition 1). They sketch the following constructive proof. Take $x \in \mathcal{X}$ and $\xi \in \mathcal{X}^{K-1}$ that satisfy their conditions for identification. Then, F_ξ and L_ξ are invertible. From Equation 12, it follows that $F_\xi^{-1} F^x_\xi = L_\xi^{-1} D_x L_\xi$. Consequently, the eigenvalues of $F_\xi^{-1} F^x_\xi$ identify D_x (up to the ordering of the K unobserved groups). With this in hand, $F_\xi^{-1} F^x_\xi L_\xi^{-1} = L_\xi^{-1} D_x$ suggests that the columns of L_ξ^{-1} can be identified with the corresponding eigenvectors of $F_\xi^{-1} F^x_\xi$. Finally, the mixing probabilities follow from $\Pi = (L'_\xi)^{-1} F_\xi L_\xi^{-1}$.

This argument is close to the classical proofs but only requires $\bar{T} = 3$ because it substitutes variation with covariates for repeated choice observations. Kasahara & Shimotsu's identification conditions ensure that there is sufficient variation in the component distributions with the covariates. They are high-level conditions that can be verified directly in the data: Kasahara & Shimotsu (2009, corollary 1) show that it is sufficient that there exist $x \in \mathcal{X}$ and $\xi \in \mathcal{X}^{K-1}$ such that F_ξ is full rank and all eigenvalues of $F_\xi^{-1} F^x_\xi$ are distinct. If either of these conditions fails, identification fails with $\bar{T} = 3$ but can be restored using their alternative result for longer panels ($\bar{T} > 3$). As they duly note, this would take us closer to the classical framework.

The data are also informative on the number K of unobserved types. Kasahara & Shimotsu present a lower bound on K that can be computed from data on two periods ($\bar{T} = 2$) only. They prove that, under additional rank conditions that cannot be empirically verified, K equals this lower bound.

2.3.3. Extensions. The baseline model requires stationarity and Markov restrictions, and independence of the observed state transition from the unobserved heterogeneity. Moreover, the assumption that Q_1 and Q_{t+1} have full support $\mathcal{X}$, almost surely, excludes that the state X_t, and thus the conditional choice probabilities, includes past choices. It also forbids monotone state processes such as Rust's (1987) bus mileage. In structural applications to dynamic discrete choice processes, these conditions may be impractical. For example, in the finite-horizon setting that is often used, we need to account for nonstationarity.

Kasahara & Shimotsu (2009) derive a range of additional results that relax one or more of these conditions, using versions of the approach for the baseline case. The main limitation of their analysis is that it does not simultaneously allow for state dependence—which they define as dependence of conditional choice probabilities on past choices—and nonstationarity.

2.4. Unobserved Markov State

A simple generalization of a permanent unobserved heterogeneity component allows this component to be updated according to a Markov process. Hu & Shum (2009) analyze the

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identification of such a generalization, applying results from the literature on nonlinear measurement error models. Like Kasahara & Shimotsu (2009), they focus on first-stage identification based on high-level assumptions and do not exploit restrictions from economic theory.

This section and the next instead focus on the identifying power of cross-equation restrictions on choice and covariate data augmented with information on choice-specific outcomes and auxiliary measurements. This approach is motivated by Heckman & Honoré's (1989) analysis of the empirical content of Roy's (1951) model of sector choice and earnings. In this static model, agents choose to work in the sector where they earn the most. In addition to sector choices A and state variables X that affect sectoral earnings, earnings in the sector where an agent works [in our notation, $u(A,X)+W(A)$] are observed. Note that, unlike Section 2.2's state variables, these earnings are directly informative on the unobservables. Heckman & Honoré provide both parametric identification results and nonparametric results that rely on large variation of sectoral earnings with the covariates.

Keane & Wolpin (1997) estimate a dynamic extension of Roy's model. The next two sections provide dynamic examples of the parametric and nonparametric approaches to the identification of such models. We first discuss Abbring & Campbell's (2005) parametric analysis of a model of firm growth, learning, and survival. Then, Section 2.5 reviews Heckman & Navarro's (2007) nonparametric analysis of a model of schooling choice and earnings.

2.4.1. Firm growth, learning, and survival. Abbring & Campbell (2005) present a model of firm growth, learning, and survival. Their main objective is to characterize the accumulation of information to entrepreneurs about their firms' profitability and to assess the entrepreneurs' effectiveness in selecting profitable firms for survival. To this end, their model distinguishes processes at three information levels. Nature draws a profit-relevant parameter from a first-order autoregressive process. The agent (entrepreneur) only observes a noisy measure of this time-varying parameter and, when deciding on firm survival, uses Bayes' rule to assess its value and predict his future profits [as in Jovanovic's (1982) model of industry evolution]. The econometrician only observes the agent's survival decisions and part of the state vector on which these are based. Thus, this model distinguishes between the agent's and the econometrician's ignorance about the state of nature. Abbring & Campbell provide identification results that show that the two can be separated. Their analysis is fully parametric but nevertheless instructive.

Firms enter at time $t = 1$. In each period t, they decide to either continue into the next period ($A_t = 0$) or to leave permanently ($A_t = 1$). A state variable N_t keeps track of survival: $N_1 = 1$ and $N_t = 1$ if and only if $N_{t-1} = 1$ and the agent decided to continue in $t-1$. If the firm survives to period $t (N_t = 1)$, it will collect period t profits

$$\varepsilon^{-1}\exp(Y_t) - \kappa_t \exp(V).$$

Here, Y_t are log sales; $\varepsilon^{-1}\exp(Y_t)$ is, under some assumptions, the firm's period t producer's surplus; and $\kappa_t \exp(V)$ are fixed production costs. If the firm is not active, it will have zero payoffs.

Log sales, and therewith profitability, are determined dynamically by a deep process W_t^* that is initialized with a draw from some normal distribution and subsequently evolves,

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independently from the firm's actions, as a first-order autoregressive normal process with linear drift. In particular, log sales satisfy

$$Y_t = W_t^* + V_t^* + Z_t + V,$$

where V_t^* and Z_t are independently and identically distributed normal disturbances, and V is a permanent normal firm-scale factor. The agent observes $W_t \equiv W_t^* + V_t^*$, Z_t, and V, but not W_t^* and V_t^* separately. That is, V_t^* is a transitory sales shock that the entrepreneur cannot tell apart from the true persistent state W_t^*. If it is nondegenerate, it will affect the agent's assessment of the true state (learning) and thus his survival decisions. In contrast, the other transitory log sales shock, Z_t, will not affect the agent's survival decisions because it is observed by the agent and unrelated to future profits. Finally, the factor $\exp(V)$ proportionally scales up profits and is known to agents. Abbring & Campbell interpret $\exp(V)$ as an entrepreneur's choice of scale upon entry, which is heterogeneous because of heterogeneity in entrepreneurial abilities.

When deciding on survival at time t, the agent knows V and the histories of N_t, Y_t, W_t, and Z_t up to and including time t; the agent also knows all the model's parameters, including the fixed cost sequence $\{\kappa_t\}$. Period t profits depend on N_t, Y_t, and V but not on the agent's current survival decision. Because of normality, the agent's expected discounted future profits only depend on the observed history through N_t, the agent's survival decision, V, the posterior mean $\hat{W}_t^* \equiv \mathbb{E}[W_t^* \mid W_1, \ldots, W_t]$ of W_t^*, and time t. Moreover, the agent can use the Kalman filter to recursively compute the posterior mean $\hat{W}_t^*$ from $\hat{W}_{t-1}^*$ and W_t. Because $\exp(V)$ simply multiplies the expected discounted profits under survival, and the payoffs under exit are zero, V does not affect the survival decision. If $N_t = 1$, the agent will survive if and only if $\hat{W}_t^*$ exceeds a threshold $\underline{w}_t$. The threshold varies with time, because the (foreseen) parameters may vary with time and because $\hat{W}_t^*$ is nonstationary under learning.

One possible specification of the agent's state is $S_t = (N_t, Y_t, \hat{W}_t^*, V)$. In terms of Section 2.1's framework,

$$u_t^*(S_t) = N_t\left[\varepsilon^{-1}\exp(Y_t) - \kappa_t \exp(V)\right].$$

The econometrician only observes A_t (and N_{t+1}) and Y_t if $N_t = 1$:

$$[(N_1 A_1, N_1 Y_1), \ldots, (N_{\bar{T}} A_{\bar{T}}, N_{\bar{T}} Y_{\bar{T}})].$$

This establishes the hierarchy of information that is key to the model's econometric implementation: The econometrician only observes a noisy measure of the agent's state, $(N_t, N_t Y_t)$. It should be clear that the state transition process does not satisfy Section 2.2's conditional independence assumption. In fact, even conditional on V, the agent's assessment $\hat{W}_t^*$ of the state of nature will help in predicting the next period's sales. Moreover, the unobservables cannot be determined from a monotonic relation with observed variables, as in Olley & Pakes (1996) and Aguirregabiria (2010). Instead, we follow a parametric identification strategy that exploits cross-equation restrictions.

2.4.2. Parametric identification using cross-equation restrictions. Clearly, from the log sales data only, we cannot tell its various components apart. However, Abbring & Campbell show that the model can be identified if we combine the sales information with choice data. In particular, they show that the parameters of the structural components of

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the log sales process, and the relevant survival thresholds $\underline{w}_1, \ldots, \underline{w}_{\bar{T}}$, can be identified if $\bar{T} \geq 3$. They provide intuition based on three ideas from the literature.

First, because survival decisions only depend on $W_t = W_t^* + V_t^*$ and not on Z_t and V, the covariance of log sales with the survival decision can be used to separate W_t, on the one hand, from Z_t and V, on the other hand. This approach is inspired by Friedman's (1957) seminal work on the permanent income hypothesis and the subsequent literature that uses the covariance of current income with consumption to disentangle permanent and transitory income shocks.

Second, once we can measure the agent's state without econometric noise, we can apply Pakes & Ericson's (1998) test for Bayesian learning and separate the true state W_t^* from the noise V_t^* in the agent's observations of it. In particular, without learning (that is, if $W_t = W_t^*$), W_t is a first-order Markov process. With learning, the full history of W_t helps predicting it.

Finally, log sales are persistent for two reasons: the structural autocorrelation of the latent process W_t^* and heterogeneity in V across firms. In the absence of selection on unobservables, the autoregression of S_2 on S_1 given $N_2 = 1$ would be linear. With selection, this autoregression involves a Heckman (1979) selection term. Abbring & Campbell argue that this selection term is informative on the relative importance of both sources of persistence because survival only depends on W_t^* and not on V.

This determines the agent's decision rules together with the parameters of the state processes, without solving his optimization problem. Abbring & Campbell complement this first-stage identification result with a second-stage analysis of the identification of the remaining primitives, using the optimization problem's structure. Under some assumptions on the time variation in the parameters, they find that the fixed cost pattern, the firm's value, and the value of the agent's exit option can be identified up to a common scale.

In contrast to most of the results reviewed here, Abbring & Campbell's analysis is firmly parametric. Their stated intuition, however, suggests that, in part, their analysis could be extended to more flexible specifications. Indeed, their proofs already apply to models that do not impose the full second-stage structure and allow for semiparametric components, such as a general nonparametric autoregression for the state process W_t. However, key aspects of their model remain parametric.

2.5. General Unobservables

In some applications, it is possible to generate lots of independent variation in, for example, payoffs across periods by varying external covariates. For instance, in their analysis of patent renewal, Pakes & Simpson (1989) entertain unobserved Markov returns to patents. They use an identification at infinity strategy, driven by external patent renewal cost variation.

Cameron & Heckman (1998) use such a strategy in their analysis of schooling choices with an ordered choice model. Cunha et al. (2007) extend their framework by allowing the thresholds that demarcate the ordered choices to depend on observed and unobserved covariates. They provide conditions under which the resulting ordered choice model can represent a dynamic optimal stopping model of schooling, with the thresholds reflecting marginal returns to schooling. Under these conditions, their identification analysis, in the spirit of Cameron & Heckman, applies to the corresponding dynamic discrete choice model. Taber (2000) analyzes a simple two-period version of a dynamic programming

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model with a very general error structure. Each of these papers exemplifies the power of the identification at infinity approach, when available.

Here we review a more recent contribution by Heckman & Navarro (2007). They extend Taber's (2000) analysis to a general finite-horizon model with information updating based on a factor model, augmented with choice-specific outcomes and auxiliary measurements.

2.5.1. Setup. Heckman & Navarro (2007) study schooling choices and their effects on earnings (see also Abbring & Heckman 2007). Time is indexed with $t \in \mathcal{T} \equiv \{1, 2, \ldots, \bar{T}\}$, for some finite time $\bar{T}$. At time 1, an agent starts in school. She receives period 1's utility from being in school and decides, based on the information available to her, whether to remain in school or leave school after the first period. If the agent is still in school at time $t \in \{2, \ldots, \bar{T}^*\}$, for some $\bar{T}^* \leq \bar{T} - 1$, she similarly collects state-dependent utility and decides whether to continue in school for another period. No more schooling is available after period $\bar{T}^* + 1$, so if the agent is still in school at time $\bar{T}^* + 1$, she has no other choice than to leave. Once the agent is out of school, by choice or by passing the highest available schooling level, she cannot return, and she collects the state-dependent utility from being out of school until time $\bar{T}$. The agent's schooling choices maximize expected discounted utility.

Denote the schooling decision at time t with A_t, where $A_t = 1$ if the agent decides to stop schooling before the next period and $A_t = 0$ if the agent remains in school. Let N_t indicate the agent's schooling state at time t. Initialize $N_1 = 0$ and set $N_{t+1} = 0$ if $N_t = 0$ and the agent decides in period t to continue in school ($A_t = 0$). If the agent has left school by period t ($N_t > 0$) or decides to leave after period t ($N_t = 0$ and $A_t = 1$), then $N_{t+1} = N_t + 1$. That is, N_t counts the periods the agent has experienced out of school, starting at 1 in the first year after leaving school. This experience will be one determinant of earnings. Note that, in period t, it can only take values between $\max\{0, t - \bar{T}^* - 1\}$ and $t - 1$.

The state at time t is $S_t = (N_t, X, W_t)$. Here, $X \equiv (X^y, X^c)$ contains state variables that are observed by both the econometrician and the agent. These may include paths of schooling cost and earnings shifters that are perfectly foreseen by the agent from time 1. The variables in W_t are state variables in the agent's information set at time t but are not known to the econometrician. We discuss their structure later.

Utility u_t^* in period t depends on the state S_t but not on the concurrent schooling choice. It equals the state-dependent earnings $y_t(S_t)$ net of the schooling costs $c_t(S_t)$:

$$u_t^*(S_t) = y_t(S_t) - c_t(S_t),$$

where

$$y_t(S_t) = \mu^y(t, N_t, X^y) + Z_t^y(N_t)$$

and

$$c_t(S_t) = \mu^c(t, X^y, X^c) + Z_t^c.$$

Here, μ^y and μ^c are deterministic functions, and $Z_t^y(n)$, for all feasible values n of N_t, and Z_t^c are mean zero shocks that are independent of X.

A key distinguishing feature of Heckman & Navarro's analysis is their specification of the agent's information structure. Note that we have already assumed that X is known to the agent from period 1. Moreover, they also make the standard assumption that all the model's parameters are known to the agent.

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It remains to specify the content of the state variables W_t that are not observed by the econometrician. For this, they specify dynamic factor models for both Z_t^y and Z_t^c and make assumptions on the way agents accumulate information on the factors and on the uniquenesses (i.e., the idiosyncratic errors in the factor model). In particular, they specify

$$Z_t^y(n) = V' v_{t,n}^y + \eta_{t,n}^y \text{ and } Z_t^c = V' v_t^c + \eta_t^c, \tag{13}$$

where V is a vector of time-invariant factors, $v_{t,n}^y$ and v_t^c are state-dependent vectors of factor loadings, and $\eta_{t,n}^y$ and η_t^c are the (zero-mean) uniquenesses. The factors and scalar components of the uniquenesses are all mutually independent, and independent of X.

Heckman & Navarro now make a set of assumptions on the agent's information structure that allows for information updating but excludes nontrivial Bayesian updating. (*a*) All the cost uniquenesses $\{\eta_t^c\}$ are known to the agent from the start. (*b*) At time t, only past and current earnings uniquenesses $\{\eta_{1,n}^y, \ldots, \eta_{t,n}^y\}$ are known to the agent. (*c*) The factors in V are sequentially and fully revealed to the agent. That is, a factor is either known or unknown at time t, and if it is known at time t, then it is known at all later times. Moreover, the factor loadings in Equation 13 are restricted so that all factors that affect $Z_t^y(n)$ and Z_t^c are known at time t. Thus, W_t includes all cost uniquenesses, all past and current earnings uniquenesses, and those factors that have been revealed by time t. Note that not all uniquenesses that the agent knows are actually relevant to her payoffs and decisions. For example, past schooling cost shocks are known but need not necessarily be included in the state W_t.

Heckman & Navarro take the information structure as given. Cunha et al. (2005) discuss identification of the information structure.

2.5.2. Nonparametric identification. Now suppose we have data on the schooling choices $(A_1, \ldots, A_{\bar{T}^\cdot})$, or, equivalently, the schooling outcomes $(N_1, \ldots, N_{\bar{T}})$; the earnings and cost shifters X; the agent's earnings $Y_t \equiv y_t(S_t)$, $t \in \{1, \ldots, \bar{T}\}$; and a vector of external measurements M. Note that the earnings data only provide information on the agent's actual earnings; we do not directly observe the counterfactual earnings $y_t(s)$ in states $s \neq S_t$.

The vector M contains variables that are not state variables or outcomes of the model but that are informative on certain aspects of the model, notably the latent factors V. For example, some of the factors could be ability, and M could be test scores that are noisy measurements of ability. Heckman & Navarro specify $M = \mu^M(X) + Z^M$ and impose a factor structure on Z^M similar to that for the earnings and cost shocks in Equation 13, again assuming full independence of the factors and the extended set of uniquenesses. Informational assumptions on the measurements are needed to ensure that they are truly external to the model and are not used by the agent to learn about the factors. Here, we simply assume that these conditions hold.

The identification analysis proceeds in two steps. First, Heckman & Navarro consider identification of the mean earnings and measurement functions μ^y and μ^M and of the distributions of the corresponding errors. Note that the measurements' regression function μ^M (and the marginal distribution of the corresponding errors) can be directly identified. Now consider identification of the means $\mu^y(t, n, \cdot)$ for an arbitrary feasible schooling outcome, that is, for the (t, n) on an arbitrary schooling path $[(1, n_1), \ldots, (\bar{T}, n_{\bar{T}})]$ in the support of $[(1, N_1), \ldots, (\bar{T}, N_{\bar{T}})]$. Heckman & Navarro effectively require that the external cost shifter X^c can be varied, independently of X^y, so that this schooling outcome occurs with probability one. This allows for the identification at infinity of the corresponding means $\mu^y(t, n, \cdot)$

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without selection bias. This can be repeated for all feasible schooling outcomes, yielding identification of $\mu^{\nu}(t, n, \cdot)$ for all feasible pairs (t, n). With these means in hand, the joint distribution of the external measurements' errors and the earnings errors corresponding to a given feasible schooling outcome follows from a similar limit argument. None of this requires the factor structure on the errors. The full joint distribution of the errors, however, can only be identified using factor analysis.

Second, they consider identification of the full structural model. Their approach is based on a standard recursive formulation of the agent's decision problem and sequentially applies identification at infinity arguments from the last decision time $\bar{T}^*$ backward to time 1. For each decision time in that sequence, this requires that they can independently vary the schooling costs in earlier periods so that the agents start in school at that time with probability one. Then, they can apply results on the nonparametric identification of static binary choice models (Matzkin 1992) to back out the parameters that appear in that period's schooling choice. Along the way, the scale of the cost functions needs to be determined, for example, by assuming that the schooling costs contain an additive component that is observed and measured on the same scale as earnings, such as tuition. Factor analysis is again used to identify joint distributions of the errors.

3. HAZARD MODELS

Hazard models naturally arise in applications in which agents make discrete decisions at random and discrete (Poisson) times (Heckman & Singer 1985, 1986). A prime example of such an application is the empirical analysis of labor market transitions using sequential job search models. Labor economists have both empirically implemented job search models as structural models (Eckstein & Van den Berg 2007) and used such models to motivate the application of MPH models (Van den Berg 2001). This section briefly reviews the current state of affairs in the identifiability of hazard models.

3.1. Hazards in Job Search and Insurance

In a basic sequential job search model, agents receive job offers at a (possibly, controlled) Poisson rate, and once a job offer arrives, they decide between accepting the offer or continuing to search without recall of the offer. The resulting hazard rate for the transition from unemployment to employment is the product of the job-offer arrival rate and the probability that a job offer is acceptable. In the simplest such setting, the job-offer arrival rate and the job-offer distribution are taken to be primitives, which may vary between agents and over time. Under some regularity conditions, the agent's decision rule is to accept any job that offers a wage above some agent-specific and time-dependent reservation wage.

Flinn & Heckman (1982) provide a seminal analysis of the identifiability of this model's primitives from unemployment duration and accepted wage data. One of their main results is that even a homogeneous and stationary version of this model is not identified, unless the class of job-offer distributions is restricted to satisfy a recoverability condition that ensures that the job-offer distribution can be uniquely determined from a left truncated version of it, the accepted wage distribution. Plainly, the effects of the job-offer arrival rate and of the job acceptance behavior on the hazard rate cannot be separated without such a condition. Later work has made some progress in resolving the recoverability problem, for example, by using equilibrium restrictions on wage-offer distributions (e.g., Eckstein & Van den Berg 2007).

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Search models have been much applied since to the empirical analysis of labor markets (reviewed in Eckstein & Van den Berg 2007). Closely related models have recently appeared in the empirical literature on asymmetric information problems in insurance. For example, in their analyses of moral hazard in car insurance, Abbring et al. (2003, 2008) specify models of an agent's driving risk and claim behavior under experience rating. In these models, insured losses arrive at a controlled Poisson rate; loss sizes are drawn from some primitive distribution; and agents only claim losses that exceed a state-dependent threshold, because of experience rating.

Progress in the analysis of the identification of such structural models driven by Poisson processes has been limited to specific applications. Here, we focus on more generally applicable insights that can be derived from the extensive parallel literature on the identifiability of the MPH model. This is an extension of Cox's (1972) proportional hazards model with a multiplicative unobserved heterogeneity factor. It was introduced by Lancaster (1979) for the analysis of state dependence and heterogeneity in unemployment duration data. The MPH model has developed since into the main tool for econometric duration analysis, but it has also been criticized for its poor structural foundations. For example, Van den Berg (2001) points out that sequential search models only lead to MPH structures under very specific assumptions, and Eckstein & Van den Berg (2007, section 3.1.7) plainly state that proportional hazards and MPH models are only good for descriptive analysis of data generated by job search models. This problem is not restricted to labor economics: Abbring et al.'s (2008) insurance model implies general interactions between individual characteristics and time.

Nevertheless, there are at least four reasons to review some key identification results for the MPH model. First, MPH results can be applied in those few specific circumstances that search models actually lead to MPH structures. Second, even if search or insurance models do not lead to proportional hazards, hazard models remain the most natural framework for the analysis of decisions taken at random discrete times. Insights from the MPH literature may help in developing identification results for nonproportional models with better structural foundations. Third, MPH results have recently been applied to an (essentially static) optimal stopping game by Honoré & de Paula (2008), and recent developments in the computation of dynamic games suggest a role for hazard models (see Section 5). Finally, the analysis of the MPH model can be applied to a completely different class of models, mixed hitting-time (MHT) models, that are closer in spirit to Section 2's discrete-time models and better embedded in economic theory. This is the subject of Section 4.

3.2. Mixed Proportional Hazards Model

We first introduce the MPH model and its implications for data on durations and covariates. Then, we briefly review the literature on its identification.

3.2.1. Setup. Suppose we have data on a continuous random time T and time-invariant covariates X. The MPH model specifies the hazard rate

$$\theta(t \mid X, V) \equiv \lim_{dt \downarrow 0} \frac{\Pr(t \leq T < t + dt \mid X, V)/dt}{\Pr(T \geq t \mid X, V)}$$

of T conditional on X and a scalar unobserved heterogeneity factor V as

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$$\theta(t \mid X, V) = \lambda(t)\phi(X)V,$$

where the baseline hazard λ: $[0,\infty) \to (0,\infty)$ has an integral $\Lambda(t) \equiv \int_0^t \lambda(\tau)d\tau$ for all finite t, ϕ: $\mathcal{X} \to (0,\infty)$ is a measurable function of the covariates, and V is distributed independently from X on $(0,\infty)$ with distribution function G. We assume that $\lim_{t\to\infty}\Lambda(t) = \infty$; with the assumption that $\phi(X)V$ is positive, this ensures that the distributions of $T|(X, V)$ and $T|X$ are nondefective [Abbring (2002) discusses identifiability of the extension with general defects, which may be of applied interest]. It follows that $\Pr(T > t \mid X, V) = \exp[-\Lambda(t)\phi(X)V]$, so that

$$\Pr(T > t \mid X) = \mathcal{L}[\Lambda(t)\phi(X)], \tag{14}$$

where $\mathcal{L}(s) \equiv \int \exp(-sv)dG(v)$, $s \in [0,\infty)$, is the Laplace transform of G. The Laplace transform $\mathcal{L}$ uniquely determines G (Feller 1971, section XIII.1, theorem 1), and we interchangeably talk about determining $\mathcal{L}$ and G in our review of the MPH model's identification.

Equation 14 expresses the data, specifically the survival function of $T|X$, in terms of the MPH model's primitives, the triplet $(\Lambda, \phi, \mathcal{L})$ (the marginal distribution of X is taken to be ancillary and is ignored in our identification discussion). We want to know whether this mapping can be inverted so that the MHT triplet (Λ, ϕ, G) can be uniquely determined from the survival function of $T|X$.

3.2.2. Identification. Following Elbers & Ridder (1982) and Ridder (1990), we focus on the two-sample case in which X takes two values, 0 and 1, only. By considering a framework with minimal regressor variation, we minimize the reliance of the identification results on such variation. Without loss of generality, we specify $\phi(X) = \beta^X$ for some $\beta \in (0,\infty)$. Note that this implicitly imposes a scale normalization $\phi(0) = 1$ on ϕ; this is innocuous because the scales of Λ and G, and therefore of θ, are free at this point. In this two-sample setup, the data provide us with two survival functions, $\bar{F}_0(t) \equiv \Pr(T > t \mid X = 0)$ and $\bar{F}_1(t) \equiv \Pr(T > t \mid X = 1)$, and the identification question is whether the two-sample MHT triplet $(\Lambda, \beta, \mathcal{L})$ is uniquely determined by $(\bar{F}_0, \bar{F}_1)$.

It should be clear upfront that we need nontrivial variation with the regressors: $\beta \neq 1$. This is a testable condition because $\bar{F}_0 = \bar{F}_1$ if and only if $\beta = 1$. If the data would instead indicate that $\beta = 1$, then there is no gain from using both $\bar{F}_0$ and $\bar{F}_1$, and the identification problem reduces to the question of whether we can identify Λ and $\mathcal{L}$ from the marginal survival function of T. This distribution satisfies $\Pr(T > t) = \bar{F}_0(t) = \bar{F}_1(t) = \mathcal{L}[\Lambda(t)]$. Clearly, for any two different Laplace transforms $\mathcal{L}$ and $\tilde{\mathcal{L}}$, we can construct observationally equivalent MHT triplets $(\Lambda, 1, \mathcal{L})$ and $(\tilde{\Lambda}, 1, \tilde{\mathcal{L}})$ by setting $\Lambda(t) = \mathcal{L}^{-1}[\Pr(T > t)]$ and $\tilde{\Lambda}(t) = \tilde{\mathcal{L}}^{-1}[\Pr(T > t)]$, $t \in [0,\infty)$. Thus, if $\beta = 1$, the MPH model is underidentified. Therefore, we assume $\beta \neq 1$ throughout.

The main results in the literature show that the two-sample MPH model is identified up to an innocuous scale normalization and a power transformation under a regularity condition on the tails of Λ and $\mathcal{L}$ (Ridder 1990, Abbring & Ridder 2009) and that full identification (up to scale) can be achieved by, in addition, fixing the behavior of one of these tails (Elbers & Ridder 1982, Heckman & Singer 1984a, Ridder & Woutersen 2003). We now review these results in some detail.

Suppose that $(\Lambda,\beta,\mathcal{L})$ and $(\tilde{\Lambda},\tilde{\beta},\tilde{\mathcal{L}})$ are two observationally equivalent MPH triplets. From Equation 14, it follows that

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$$\mathcal{L}\circ(\beta\mathcal{L}^{-1}) = \bar{F}_1\circ(\bar{F}_0^{-1}) = \tilde{\mathcal{L}}\circ(\tilde{\beta}\tilde{\mathcal{L}}^{-1}), \tag{15}$$

where $\circ$ denotes function composition. Composing both sides of Equation 15 n times with themselves yields $\mathcal{L}\circ(\beta^n\mathcal{L}^{-1}) = \tilde{\mathcal{L}}\circ(\tilde{\beta}^n\tilde{\mathcal{L}}^{-1})$. Moreover, we can repeat this analysis with the role of both samples exchanged and derive a similar relation for negative n. Thus, we have

$$\mathcal{L}\circ(\beta^n\mathcal{L}^{-1}) = \tilde{\mathcal{L}}\circ(\tilde{\beta}^n\tilde{\mathcal{L}}^{-1}),\ n\in\mathbb{Z}. \tag{16}$$

Now define $\mathcal{K} \equiv \tilde{\mathcal{L}}^{-1}\circ\mathcal{L}$. Observational equivalence, specifically $\tilde{\mathcal{L}}\circ\tilde{\Lambda} = \mathcal{L}\circ\Lambda$, implies that also $\mathcal{K} = \tilde{\Lambda}\circ\Lambda^{-1}$. Consequently, $(\Lambda,\beta,\mathcal{L}) = (\tilde{\Lambda},\tilde{\beta},\tilde{\mathcal{L}})$ if and only if $\mathcal{K}(s) = s$, $s\in[0,\infty)$. More generally, if we can determine $\mathcal{K}$, perhaps up to a few free parameters, then we can characterize the relation between any two observationally equivalent MHT triplets.

From Equation 16, it follows that $\tilde{\beta}^n\mathcal{K}(s) = \mathcal{K}(\beta^n s)$ and $\tilde{\beta}^n\mathcal{K}'(s) = \beta^n\mathcal{K}'(\beta^n s)$, so that

$$\frac{\mathcal{K}'(s)}{\mathcal{K}(s)} = \frac{\mathcal{K}'(\beta^n s)}{\mathcal{K}(\beta^n s)/\beta^n};\ \text{and } n\in\mathbb{Z}, s\in(0,\infty). \tag{17}$$

The limiting behavior of the right-hand side of Equation 17 as $|n|\to\infty$ is key to all identification results in the MPH literature. Because $\mathcal{K} = \tilde{\mathcal{L}}^{-1}\circ\mathcal{L} = \tilde{\Lambda}\circ\Lambda^{-1}$, this limiting behavior can be controlled by making assumptions on the tails of $\mathcal{L}$ (and $\tilde{\mathcal{L}}$) and Λ ($\tilde{\Lambda}$). Specifically, Abbring & Ridder (2009) show that, under a regularity condition on these tails, the right-hand side converges to ω / s, for some $\omega\in(0,\infty)$. Consequently, $\mathcal{K}(s) = \delta s^{\omega}$, for some $\delta\in(0,\infty)$. This establishes Ridder's (1990) theorem 1 for the MPH model: Any two observationally equivalent MHT triplets $(\Lambda,\beta,\mathcal{L})$ and $(\tilde{\Lambda},\tilde{\beta},\tilde{\mathcal{L}})$ satisfy

$$\mathcal{L}(s) = \tilde{\mathcal{L}}(\delta s^{\omega}), s\in[0,\infty); \tilde{\Lambda} = \delta\Lambda^{\omega};\ \text{and } \tilde{\beta} = \beta^{\omega};\ \text{for some } \omega,\delta\in(0,\infty).$$

The parameter δ indexes an innocuous reassignment of the hazard's scale between the baseline hazard and the unobserved heterogeneity factor. The parameter ω, however, indexes a power transformation with substantial meaning. For example, with a linear integrated baseline $\Lambda(t) = t$, we only know that $\tilde{\Lambda}(t) = \delta t^{\omega}$ is Weibull. This implies that we cannot tell whether the baseline hazard increases ($\omega > 1$) or decreases ($\omega < 1$); that is, we cannot even sign duration dependence of the agent's hazard.

This motivates making further assumptions that narrow the class of observationally equivalent models. The literature has invariably strengthened Abbring & Ridder's (2009) regularity condition, either by making further assumptions on the right tail of the heterogeneity distribution (Elbers & Ridder 1982, Heckman & Singer 1984a) or by restricting the behavior of the baseline hazard near zero (Ridder & Woutersen 2003). We review the two simplest cases. Consider again two observationally equivalent MPH triplets (Λ, β, $\mathcal{L}$) and $(\tilde{\Lambda},\tilde{\beta},\tilde{\mathcal{L}})$.

In one case, Elbers & Ridder assume that V has a finite mean. Because the mean of a random variable, if it exists, equals minus the derivative of its Laplace transform at zero, this implies that both $0 < -\mathcal{L}'(0+) < \infty$ and $0 < -\tilde{\mathcal{L}}'(0+) < \infty$, where $\mathcal{L}'(0+) \equiv \lim_{s\downarrow 0}\mathcal{L}'(s)$, and so on. From the definition of $\mathcal{K}$, because $\mathcal{K}(0+) = 0$, this gives $0 < \mathcal{K}'(0+) = \lim_{s\downarrow 0}\mathcal{L}'(s)/\tilde{\mathcal{L}}'[\mathcal{K}(s)] < \infty$. Consequently, the right-hand side of Equation 17 converges to $\mathcal{K}'(0+)/[s\mathcal{K}'(0+)] = 1/s$, so that Ridder's (1990) theorem 1 holds with $\omega = 1$.

In another case, Ridder & Woutersen instead assume that the baseline hazard is bounded away from zero and ∞ at zero. Because $\Lambda^{-1}(0+) = 0$, this implies that $0 < \mathcal{K}'(0+) = \lim_{s\downarrow 0}\tilde{\Lambda}'[\Lambda^{-1}(s)]/\Lambda'[\Lambda^{-1}(s)] < \infty$. Consequently, the argument for Elbers & Ridder's case applies.

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Honoré (1993) shows that much of the MPH model's restrictive assumptions can be relaxed if stratified data are available, with the multiple observations in each stratum known to share the same unobserved heterogeneity factor. Similarly, Heckman & Taber (1994) show that time-varying covariates may help in relaxing some of the model's assumptions.

4. HITTING-TIME MODELS

MHT models specify durations as the first times some latent process hits a threshold that may depend both on observed covariates and unobserved heterogeneity. They imply conditional hazard rates, but, unlike Section 3's models, they do not specify these directly. Hitting-time models have been applied to strike durations (Lancaster 1972) and have inspired the analysis of unemployment durations (Shimer 2008). They have recently gained popularity in the statistics literature (e.g., Lee & Whitmore 2006) and are closely related to real options models of investment timing and their multiple-state generalizations. Possible applications in fields other than labor economics include marriage and divorce, firm entry and exit, and credit default.

This section discusses recent advances in the identification analysis of a class of MHT models that explicitly build on Section 3's results from the MPH literature. We first sketch the structural background.

4.1. Hitting Times and Optimal Stopping

Hitting times, based on threshold optimal decision rules, naturally arise in economic models in which agents optimally time discrete actions, with payoffs driven by Brownian motion (Dixit & Pindyck 1994, Stokey 2009) or more general processes (Kyprianou 2006, Boyarchenko & Levensdorskiĭ 2007).

For example, in McDonald & Siegel's (1986) real options model of investment timing, agents are endowed with an option to invest in a project, at a time of their choice. Investment incurs a given cost; in return, the agent receives the project's value at the time of the investment. The log of this value follows a Brownian motion. The agent maximizes his expected discounted payoffs by investing when the project's value hits a time-invariant threshold. Primitive heterogeneity (e.g., variation in initial project values, investment costs, and discount rates across agents) induces heterogeneity in the threshold. Consequently, data on investment times and covariates can be analyzed with an MHT model.

To provide more insight into the way primitive heterogeneity may generate threshold heterogeneity, and thus an MHT structure, we develop a well-known multiple-state version of this model in somewhat more detail. Following Abbring (2007), we adapt Dixit's (1989) model of firm entry and exit to the analysis of unemployment durations.

Consider a labor market in which workers continuously choose between unemployment and employment. A worker earns a flow b when unemployed and $u_t \equiv u_0 \exp[\mu t + \sigma B_t]$ when employed at calendar time t, where $\{B_t\}$ is a standard Brownian motion. Note that u_t is a geometric Brownian motion with drift, and $\mathbb{E}[u_t] = u_0 \exp[(\mu + \sigma^2 / 2)t]$. Workers incur a lump sum cost $\underline{k} \geq 0$ when they leave their job and pay $\bar{k} \geq 0$ when they enter a job. They maximize expected earnings, discounted at a rate $\rho > \mu + \sigma^2 / 2$. From Dixit's analysis, it follows that an unemployed worker enters employment when u_t increases above $\bar{u}$ and resigns when u_t falls below $\underline{u}$, where $\bar{u} = \underline{u}$ if $\bar{k} = \underline{k} = 0$, and $\bar{u} > \underline{u}$ otherwise.

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An MHT model can be applied to an inflow sample of unemployment durations. Normalize the start time of each unemployment spell in the sample to zero. Then, the unemployed start the sampled spell with earnings $u_0 = \underline{u}$ and end their spell when earnings hit the exit threshold $\bar{u} \geq \underline{u}$. Define $W_t \equiv \ln u_t - \ln \underline{u}$ and note that $\{W_t\}$ is a Brownian motion with drift term μt. Then, we can equivalently say that workers initially have normalized log earnings $W_0 = 0$ and leave for employment when $\{W_t\}$ hits $\underline{w} \equiv \ln \bar{u} - \ln \underline{u}$. From Dixit's (1989) analysis, it follows that $\underline{w}$ varies on $[0,\infty)$ with observed and unobserved determinants of $\bar{k}$ and $\underline{k}$, with $\underline{w} = 0$ only in the frictionless limit. Thus, a loglinear specification of the threshold $\underline{w}$ in terms of the observed and unobserved heterogeneity factors is natural.

Here, we study whether such a model is identified. If it would be, its empirical analysis would yield estimates of the latent earnings process and the agents' decision rules. These could in turn be used as inputs in a further analysis of the model's remaining primitives. Thus, we first focus on first-stage identification. Because there are few characterizations of optimal decision rules for general preference structures that can be put to use in a second-stage identification analysis, such an analysis needs to proceed on a case-by-case basis. We give one explicit example at the end of this section.

4.2. Mixed Hitting-Time Model

We first introduce the MHT model's primitives. Then, we characterize its implications for data on durations and covariates and review some results on its identification.

4.2.1. Setup. Suppose we have data on random durations T. These durations could be, for example, investment times, strike durations, marriage durations, or credit default times. We also observe a vector of covariates X, which contains external shifters of those durations. An MHT model specifies T as the first time that a latent scalar process $\{W_t; t \in [0,\infty)\}$ crosses a threshold that depends on X and some unobservables V. We assume throughout that W_t is not observed by the econometrician; we discuss the use of time-varying covariates in this framework in Section 5. Thus, MHT models combine features of Section 2.3's mixture models and Section 2.4's models with unobserved states.

Let T_w denote the first time that the latent process $\{W_t\}$ exceeds a threshold $w \in [0,\infty)$. A (proportional) MHT model specifies that T is the first time that $\{W_t\}$ crosses $\phi(X)V$, or

$$T = T_{\phi(X)V},$$

for some positive measurable function ϕ and positive random variable V, with (X,V) independent of $\{W_t\}$. The hitting times T_w characterize durations for given thresholds $w \in [0, \infty)$ and thus for given individual characteristics (X,V). Variation in $\phi(X)V$ corresponds to heterogeneity in individual thresholds. We assume that V is independent of X.

For example, in Section 4.1's application of Dixit's (1989) model to unemployment durations, $\{W_t\}$ is a Brownian motion with drift that represents normalized earnings in a job. An agent with given transition costs will leave unemployment if these normalized earnings exceed a given threshold $\underline{w}$. His unemployment durations $T_{\underline{w}}$ will have an inverse Gaussian distribution, and, conditional on X, the observed unemployment durations T will have a mixed inverse Gaussian distribution.

4.2.2. Characterization for the Lévy case. Abbring (2007) characterizes the MHT model for the case in which $\{W_t\}$ is a spectrally negative Lévy process. A Lévy process is

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a process with stationary and independent increments. An important example of a Lévy process is the scalar Brownian motion with drift, which is the only Lévy process with continuous sample paths. In general, Lévy processes may have jumps. Key examples of pure jump Lévy processes are compound Poisson processes, which have independently and identically distributed jumps at Poisson times. From an econometric perspective, a Lévy process $\{W_t\}$ is a semiparametric object that is fully characterized by a drift parameter, the dispersion parameter of its Brownian motion component, and the characteristic measure of its jump process.

A spectrally negative Lévy process is a Lévy process that has no positive jumps. Spectrally negative Lévy processes are much easier to analyze than general Lévy processes. In particular, because they cannot jump across a given positive threshold, under standard regularity conditions, they attain the threshold at the first hitting time: $W_{T_w} = w$ if $T_w < \infty$. In turn, this ensures that the hitting-time process $\{T_w\}$ can be analyzed as in the Brownian motion case, which is a special case of a spectrally negative Lévy process.

In particular, Abbring (2007) shows that Laplace transform of the distribution of T_w satisfies

$$\mathcal{L}_{T_w}(s) = \exp[-\Lambda(s)w].$$

The characteristic exponent Λ of the hitting-time process is determined by the parameters of the latent process $\{W_t\}$: its drift parameter, the dispersion parameter of its Brownian motion component, and the characteristic measure of its jump component. It follows that the Laplace transform of the distribution of T conditional on (X, V) is given by

$$\mathcal{L}_T(s \mid X, V) = \exp[-\Lambda(s)\phi(X)V].$$

Consequently, the Laplace transform of the distribution of T conditional on X equals

$$\mathcal{L}_T(s \mid X) = \mathcal{L}[\Lambda(s)\phi(X)], \tag{18}$$

where $\mathcal{L}$ is the Laplace transform of the distribution of V.

Equation 18 expresses an object that can be estimated from data—recall that, for given s, $\mathcal{L}_T(s|X)$ is simply a moment of the distribution of $T|X$—in terms of the model primitives: $\mathcal{L}$, Λ, and ϕ. The identification question is whether this mapping can be inverted, that is, whether we can uniquely determine the MHT triplet $(\Lambda, \phi, \mathcal{L})$ from data on $\mathcal{L}_T(\cdot|X)$. If we can, in turn, $\mathcal{L}$ uniquely determines the distribution of the unobserved heterogeneity factor V, and the characteristic exponent Λ fully characterizes the distributional properties of the hitting-time process $\{T_w\}$ and of the latent Lévy process $\{W_t\}$.

4.2.3. Identification. The key insight in Abbring (2007) is that the expression for the Laplace transform of the distribution of $T|X$ (Equation 18) has the same functional form as the expression for its survival function in the MPH model (Equation 14). That is, the MHT and MPH models are not substantially the same—after all, completely different representations of the data satisfy the same expression—but their identification can be analyzed in parallel.

Consider the special case that $\{W_t\}$ is a Brownian motion with drift parameter $\mu \in [0, \infty)$ and dispersion parameter $\sigma \in (0, \infty)$. Then, Equation 18 holds, with

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$$\Lambda(s) = \frac{\sqrt{\mu^2 + 2\sigma^2 s} - \mu}{\sigma^2}.$$

Note that this exponent Λ is indeed increasing, with $\Lambda(0) = 0$ and $\lim_{s\to\infty}\Lambda(s) = \infty$. Consequently, it has the main properties of an integrated baseline hazard in the MPH model, and identification results for that model directly apply.

In particular, suppose that $\phi(X) = \exp(X'\beta)$ for some parameter vector β. Note that Λ is differentiable on $(0, \infty)$ and that $0 < \lim_{s\downarrow 0}\Lambda'(s) = \mu^{-1} < \infty$. Thus, Ridder & Woutersen's (2003) identification result for the MPH model (see Section 3.2.2) implies that μ, σ, β, and $\mathcal{L}$ are uniquely determined from $\mathcal{L}_T(\cdot|X)$ under support conditions on the covariates X, up to two scale normalizations.

Abbring (2007) develops this and many other results for the general case in which $\{W_t\}$ is a spectrally negative Lévy process, along the lines of the results reviewed in Section 3. Some distinguishing features of his analysis for the MHT framework are the following. First, defects naturally arise in the MHT framework. There may be stayers if a subpopulation entertains an infinitely high threshold and never leaves. In addition, among those with finite thresholds, some may never move. This, for example, happens if W_t is a Brownian motion with negative drift. The distinction between stayers and defecting movers is often of substantial interest (Abbring 2002), and Abbring (2007) allows for both. Second, the special structure on Λ facilitates sharper identification results and tighter interpretation of these results than is possible in the MPH model. Third, in some cases, the optimal stopping problems that underlie the MHT structure imply conditions for point identification like Elbers & Ridder's (1982) finite mean assumption on V. For example, in Section 4.1's application of Dixit's (1989) model to unemployment durations, unbounded primitive heterogeneity in the primitive costs of moving back into unemployment only translates into bounded threshold heterogeneity.

4.3. Identification of the Primitives of Optimal Stopping

Section 4.2's identification results for the MHT model can be used as input in a second-stage identification analysis of the primitives of an optimal stopping model that reduces to the MHT model. Such a second-stage analysis needs to be tailored to the application at hand. Here, we present an example in which agents solve Kyprianou & Surya's (2005) extension of Novikov & Shiryaev's (2005) investment timing problem to the Lévy case.

Agents choose the best time to invest an amount $\bar{k}$ in a project that returns $u_0 + W_t$ when they invest at time t, where u_0 is the project's initial value and $\{W_t\}$ is a Lévy process. They discount future payoffs at a rate $\rho > 0$ and receive zero payoffs if they never invest. Abbring (2007) notes that, with $\{W_t\}$ spectrally negative, the agent will invest when $\{W_t\}$ first hits the threshold $\max\{\bar{k} - u_0 + \Lambda(\rho)^{-1}, 0\}$.

This problem fits the MHT framework, with heterogeneity in $\bar{k} - u_0$ and ρ inducing threshold heterogeneity. An analysis with the MHT model determines $\delta^{-1}\Lambda$—that is, the distribution of δW_t—and the distribution of δV, with $\delta \in (0, \infty)$ an unknown scale factor. For expositional convenience, we suppress covariates X, so that the threshold simply equals V.

First, suppose that ρ and $\bar{k} - u_0$ vary in the population such that $\bar{k} - u_0 + \Lambda(\rho)^{-1}$ has nonnegative support, so that the threshold $V = \bar{k} - u_0 + \Lambda(\rho)^{-1}$. Then, from the distribution of δV, we can determine the distribution of $\delta[\bar{k} - u_0] + \delta/\Lambda(\rho)$. If ρ is fixed and

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known, then $\delta / \Lambda(\rho)$ is known, and the distribution of $\bar{k} - u_0$ is identified up to scale. If ρ is fixed but unknown, this distribution is only identified up to location and scale. Similarly, if $u_0 - \bar{k}$ is homogeneous and ρ is heterogeneous, the monotonicity of Λ ensures that we can identify the distribution of ρ up to the threshold's location and scale.

In general, there may be a mass of agents who choose zero thresholds and invest immediately. These are agents with high net initial project values $u_0 - \bar{k}$ and high discount rates ρ. If the share of agents who invest immediately is observed, censored distributions of primitives can be identified; otherwise, only truncated distributions can be determined.

One may wonder how agents end up with an investment option that yields immediate profit in the first place. The simple investment timing model presented here treats u_0 as a free parameter and does not address this question. A two-state model such as Dixit's (1989) in Section 4.1 specifies both the timing of the investment of interest and the way agents end up with the option to invest. Such a model ensures that primitive heterogeneity translates into positive thresholds.

5. CONCLUSION

The research reviewed in this article has much enriched our understanding of the identification of dynamic discrete choice models. Important advances have been made in the identification analysis of such models with unobserved heterogeneity and state variables. Many open questions remain; I list a select few. (*a*) Both Section 2.3's discrete-time analysis and Section 4.2's continuous-time analysis focus on first-stage identification of decision rules and state processes. Some exploration of primitive conditions for Section 2.3's high-level identification assumptions, in particular, would be useful. Moreover, the second-stage identification of economic primitives using Section 4.2's hitting-time framework needs to be further developed, along the lines of Section 4.3. (*b*) Section 2.4 suggests that semiparametric identification results that do not rely on identification at infinity can be developed for specific models with unobserved state variables. (*c*) Section 3 motivates the analysis of nonproportional hazard models that are structured to better fit in with dynamic economic theory. (*d*) The hitting-time framework needs be augmented with time-varying covariates in a way that is consistent with theory. One option would be to model such covariates as noisy measurements of the latent state, as in Section 2.4's discrete-time models.

Dynamic discrete games, such as firm entry and exit games in industrial organization, are an important next step in the analysis of discrete decision problems. Most research has focused on Markov perfect equilibria in discrete-time games (see Ackerberg et al. 2007 for a review). In such equilibria, each agent solves a dynamic program given the other players' strategies. Consequently, much of Section 2's identification results for single-agent dynamic discrete choice processes in discrete time can be put to good use in the analysis of such games (e.g., Bajari et al. 2009).

There are few results on the identification of continuous-time games. One exception is Honoré & de Paula's (2008) identification analysis of an optimal stopping game. They apply Section 3's results for hazard models to what is, essentially, a static timing game. Recent work on the analysis of Markov perfect equilibria in truly dynamic, continuous-time games suggests that there may be substantial computational benefits from using econometric models based on such games (Doraszelski & Judd 2005). As in the discrete-time case, identification results for continuous-time single-agent models will be of use in the identification analysis of such models.

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DISCLOSURE STATEMENT

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